

INPUT AND OUTPUT SYSTEMS, SENSORS, AND SIGNAL ACQUISITION

Push Button



Used to make or break a circuit.

Temperature Sensor



Measures temperature.

Camera



Captures images or video.

GPS Module



Provides position and location.

Ultrasonic Sensor



Measures distance using sound waves.

LED



Emits light when powered.

Motor



Converts electrical energy to motion.

LCD Display



Displays text or characters.

Relay



Acts as an electrically controlled switch.

Speaker



Produces sound.

Analog Sensor



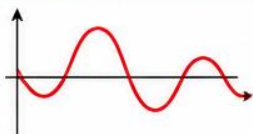
Outputs a continuous analog signal.

Potentiometer



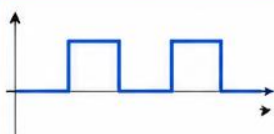
Variable resistor for analog control.

Analog Signal



Continuous signal with varying values.

Digital Signal



Discrete signal with two levels: 0 or 1.

LM35 Temperature Sensor



Outputs voltage proportional to temperature.

DHT11



Measures temperature and humidity.

Ultrasonic Sensor (HC-SR04)



Measures distance using ultrasound.

GPS Module (NEO-6M)



Provides location, speed and time data.

Thermocouple



Measures temperature using two dissimilar metals.

Piezoelectric Sensor



Generates voltage when mechanical stress is applied.

Pressure Sensor



Measures pressure of gases or liquids.

Distance Sensor



Measures distance between objects.

Light Sensor



Detects light intensity.

Motion Sensor



Detects human or object movement.

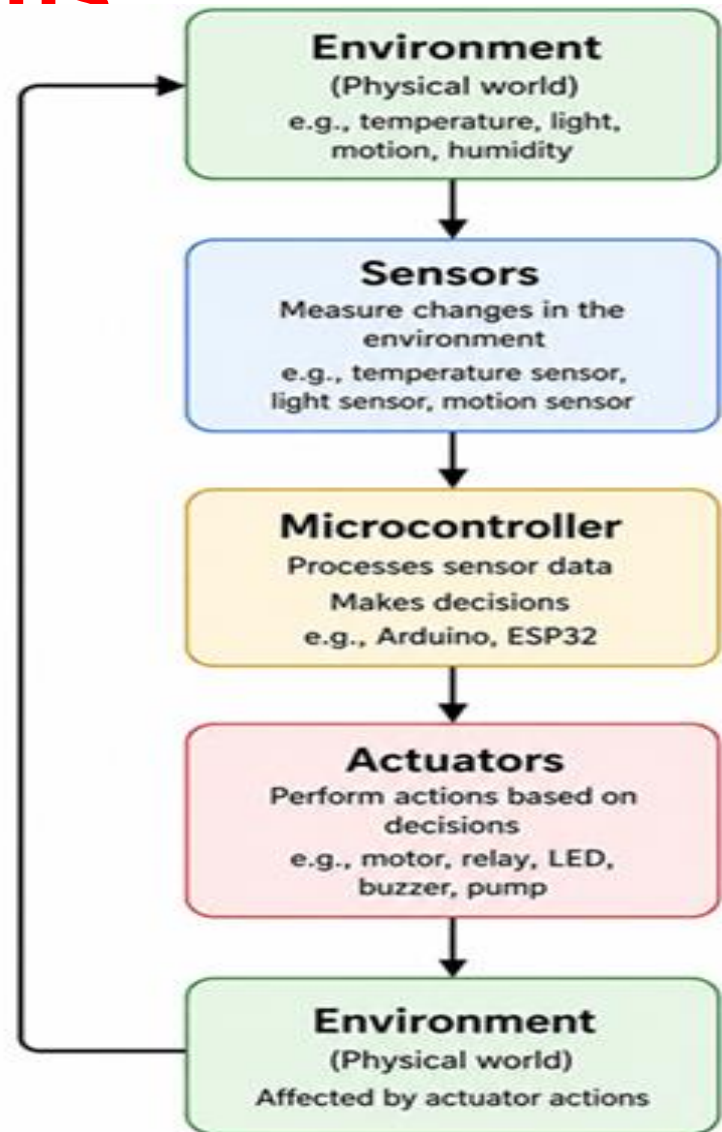
Acceleration Sensor



Measures acceleration and tilt.

Introduction to Input and Output Systems

- An embedded system interacts with its environment through input and output devices.



General Embedded System Structure

Inputs

- Inputs provide information to the system.
- Examples:
 - Push buttons
 - Temperature sensors
 - Cameras
 - GPS modules
 - Ultrasonic sensors

Types of Sensors and Input Devices

Push buttons



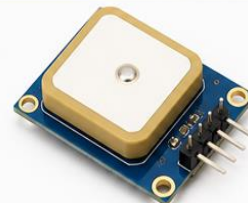
Temperature sensors



Cameras



GPS modules



Ultrasonic sensors



Types of Sensors and Input Devices

Push buttons



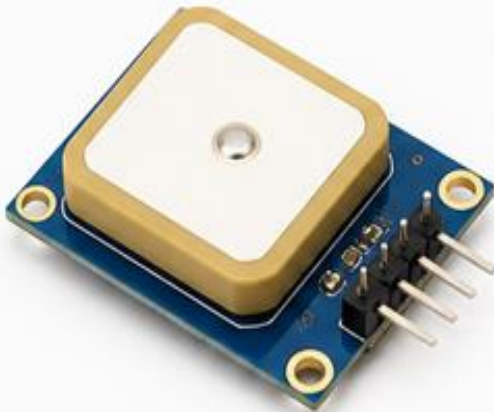
Temperature sensors



Cameras



GPS modules



Ultrasonic sensors

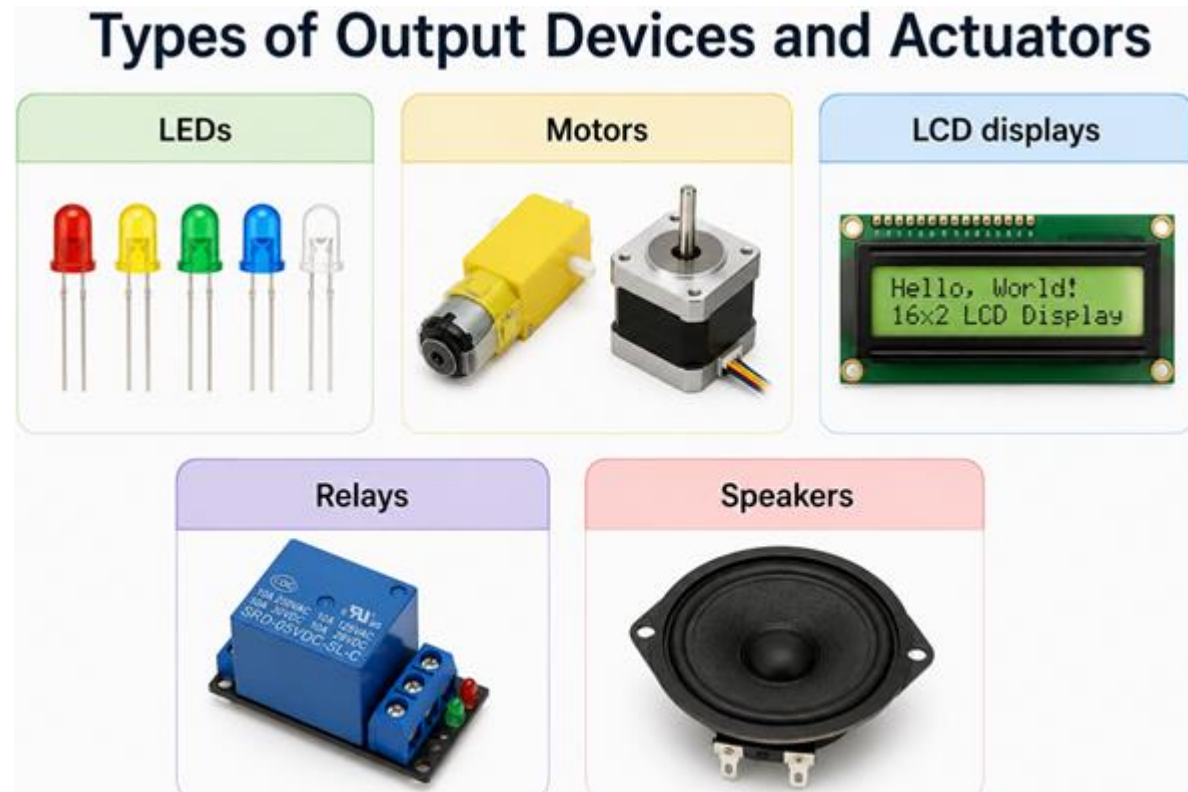


Outputs

- Outputs perform actions.

- Examples:

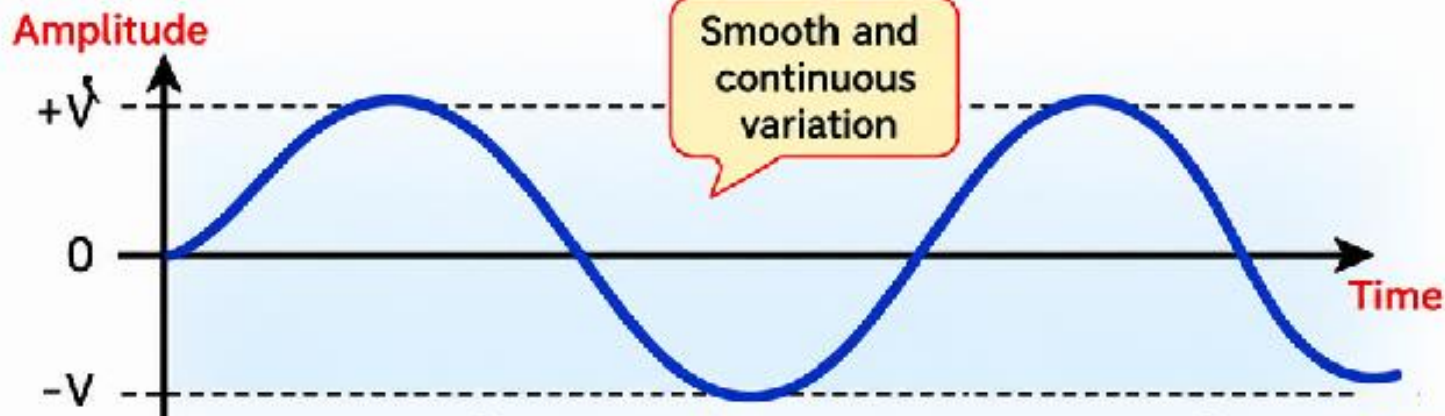
- LEDs
- Motors
- LCD displays
- Relays
- Speakers



Analog Signals

- Analog signals vary continuously.
- Example:
 - Temperature changes continuously.

An analog signal is a continuous signal that varies smoothly over time

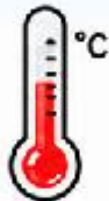


Examples

- ✓ Sound waves
- ✓ Temperature
- ✓ Light intensity
- ✓ Pressure
- ✓ Voltage from microphone



Sound



Temperature



Light



Pressure



Microphone
output voltage

The amplitude changes continuously with time and can take any value between $+V$ and $-V$.

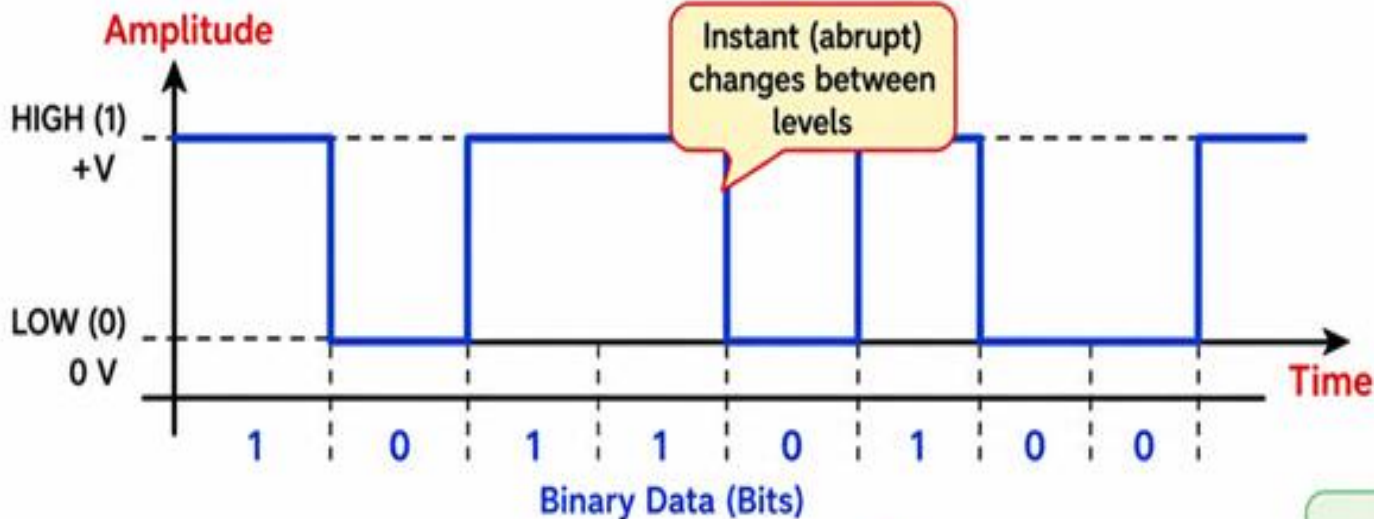
Characteristics of Analog Signals

- Infinite values
- Continuous
- Examples:
 - Sound
 - Temperature
 - Light intensity

Digital Signals

- Digital signals have discrete levels

A digital signal has **discrete (stepped) levels** and changes between them **instantly**.

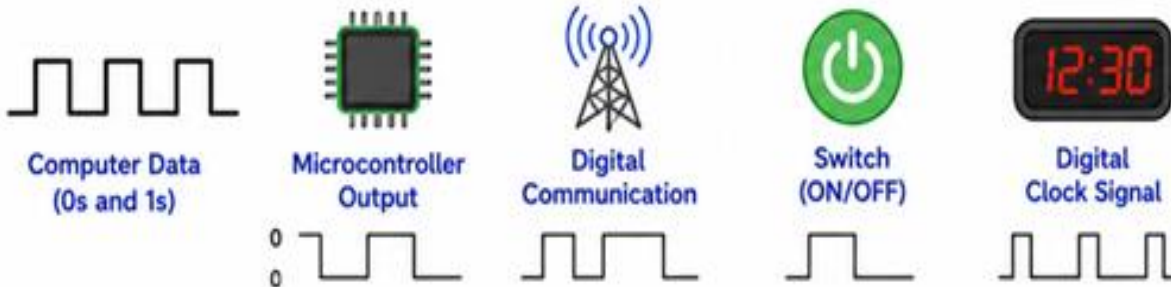


Examples

- ✓ Computer data (0s and 1s)
- ✓ Digital clock signals
- ✓ Microcontroller outputs
- ✓ Communication data
- ✓ Switch (ON/OFF) states

Key Features

- ✓ Takes on discrete values (usually 0 or 1)
- ✓ Has a finite number of levels
- ✓ Changes between levels instantly
- ✓ More resistant to noise
- ✓ Used in digital electronics and computers



Characteristics of Digital Signals

- Binary values
- 0 or 1
- Examples:
 - Switches
 - Digital sensors
 - Logic circuits

Comparison of Analog and Digital Signals

Feature	Analog	Digital
Nature	Continuous	Discrete
Accuracy	High	Limited by resolution
Noise Immunity	Low	High
Processing	Difficult	Easy
Storage	Difficult	Easy

Sensors

- A sensor is a device that detects physical phenomena and converts them into electrical signals.
- Examples of physical phenomena:
 - Temperature
 - Pressure
 - Distance
 - Light
 - Motion
 - Acceleration

Common Physical Quantities and Their Sensors



Temperature

Measures how hot or cold something is.

Sensor Examples:

LM35, DS18B20, Thermistor



Applications:

Weather monitoring, HVAC, industrial processes



Pressure

Measures the force exerted per unit area.

Sensor Examples:

BMP180, MPX5700AP, Pressure Transducer



Applications:

Weather monitoring, Altitude measurement, Industrial control



Distance

Measures the space between objects.

Sensor Examples:

HC-SR04 (Ultrasonic), LiDAR, ToF Sensor



Applications:

Obstacle detection, Robotics, Parking assistance



Light

Measures the intensity of light.

Sensor Examples:

LDR (Photoresistor), BH1750, TSL2561



Applications:

Automatic lighting, Display brightness control, Solar tracking



Motion

Detects movement or change in position.

Sensor Examples:

PIR Sensor, IR Sensor, Gyroscope



Applications:

Security systems, Automatic doors, Energy saving



Acceleration

Measures change in velocity or direction.

Sensor Examples:

ADXL345, MPU6050, Accelerometer



Applications:

Fall detection, Vibration analysis, Gesture recognition

Sensor Classification

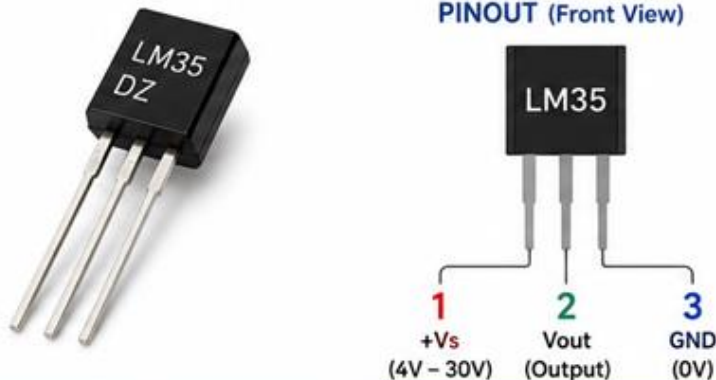
- Based on Signal Type
 - Analog Sensors
 - Digital Sensors
- Based on Energy Source
 - Active Sensors
 - Passive Sensors

Based on Signal Type: Analog Sensors

- Produce continuous signals.
- Examples:
 - LM35 temperature sensor
 - Potentiometer

LM35 TEMPERATURE SENSOR

The LM35 is a precision integrated circuit temperature sensor whose output voltage is linearly proportional to the Celsius (Centigrade) temperature.



POTENTIOMETER

A potentiometer is a variable resistor with three terminals. It produces an adjustable output voltage (voltage divider).



TERMINALS

- 1 (End 1) : Connected to one end of the resistive element
- 2 (Wiper) : Adjustable output (tap)
- 3 (End 2) : Connected to the other end of the resistive element

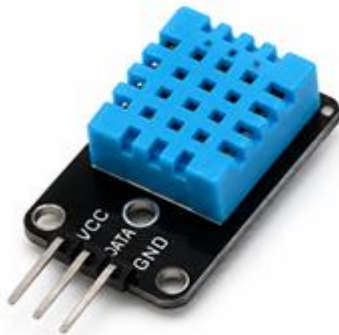
SYMBOL



Based on Signal Type: Digital Sensors

- Produce digital outputs.
- Examples:
 - DHT11
 - Ultrasonic sensor
 - GPS

DHT11



Measures temperature and humidity.

Ultrasonic Sensor (HC-SR04)



Measures distance using ultrasonic waves.

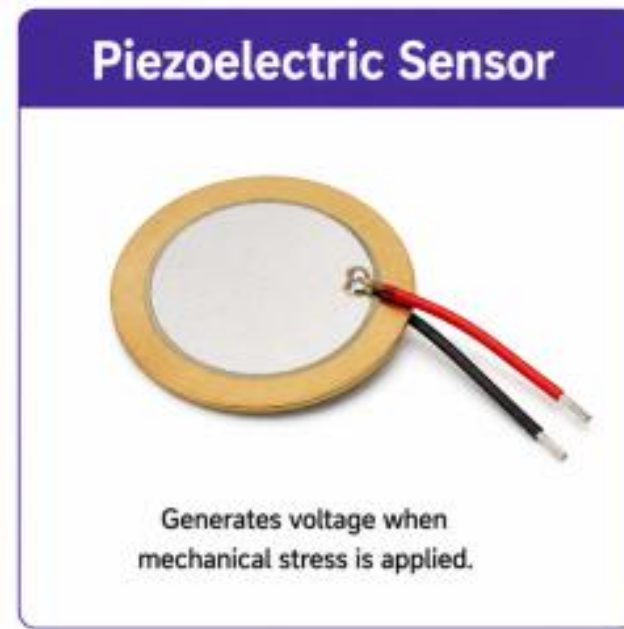
GPS Module (NEO-6M)



Provides location, speed and time data.

Based on Energy Source: Active Sensors

- Generate their own signals.
- Examples:
 - Thermocouple
 - Piezoelectric sensor



Based on Energy Source: Passive Sensors

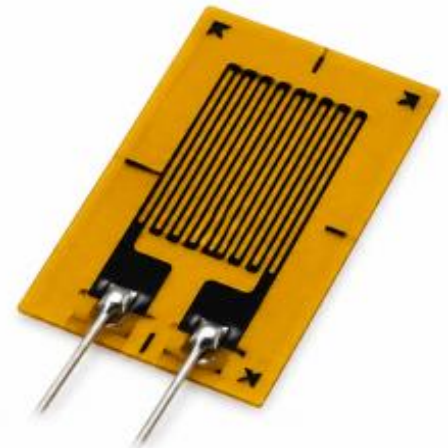
- Require external excitation.
- Examples:
 - LDR
 - Strain gauge

LDR (Light Dependent Resistor)



Resistance changes with light intensity.

Strain Gauge



Resistance changes with applied strain or deformation.

Sensor Characteristics

- Several parameters determine sensor performance.
 - **Sensitivity**

Sensitivity

- Measures how much output changes for a change in input
- Mathematically:

$$\textit{Sensitivity} = \frac{\Delta \textit{Output}}{\Delta \textit{Input}}$$

- Example:
 - Temperature sensor: 10 mV per °C

Accuracy

- Indicates closeness to true value.
- Example:
 - True Temperature = **30°C**
 - Sensor Reading = **29.8°C**
- Error:

$$\begin{aligned} \text{Error} &= 30 - 29.8 \\ &= 0.2^{\circ}\text{C} \end{aligned}$$

Precision

- Ability to reproduce measurements consistently

Resolution

- Smallest detectable change
- Example:
 - ADC resolution = 0.01V
-

Range

- Minimum and maximum measurable values.
- Example:
 - $0^{\circ}\text{C} - 100^{\circ}\text{C}$

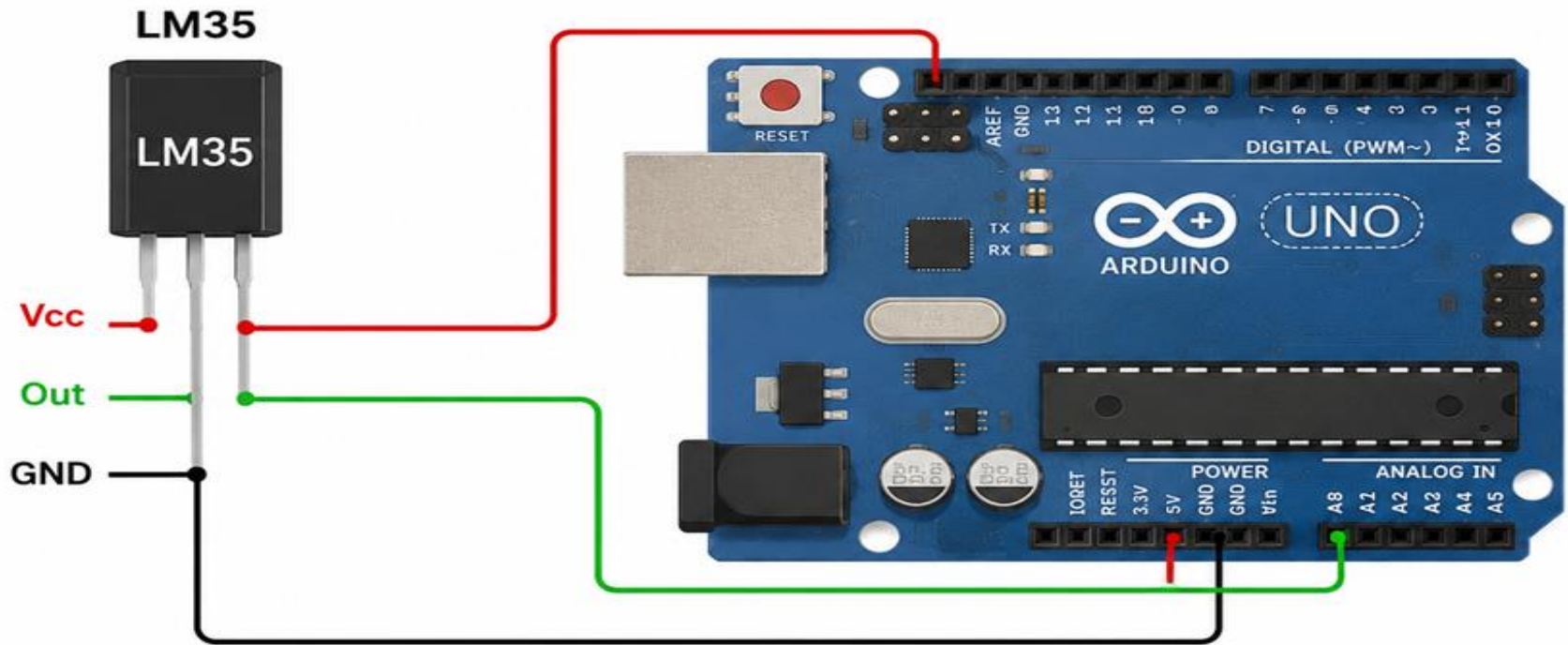
Temperature Sensors

- Temperature sensors are widely used in embedded systems and robots.
- Applications:
 - Industrial automation
 - Weather stations
 - Smart homes
 - Robot protection systems

LM35 Temperature Sensor

- Features:
 - Analog output
 - Linear response
- Relationship:
 - $Output = 10mV/^{\circ}C$
- **Example**
 - Temperature = $30^{\circ}C$
 - Output: $30 \times 10mV = 300mV$
-

LM35 Interfacing



LM35 Pin	Name	Function
1	Vcc	Supply Voltage (4V – 30V)
2	Out	Analog Output (10 mV / °C)
3	GND	Ground (0V)

Output Voltage (Vout)

$$V_{out} = 10 \text{ mV} \times \text{Temperature } (^{\circ}\text{C})$$

$$\text{Temperature } (^{\circ}\text{C}) = V_{out} / 0.01$$

Example:

If temperature = 25°C

$$V_{out} = 10 \text{ mV} \times 25 = 250 \text{ mV (0.25 V)}$$

Arduino will read ~ 0.25 V on A0

Arduino Program for LM35 Sensor

```
int sensorPin=A0;

void setup()
{
  Serial.begin(9600);
}

void loop()
{
  int value=analogRead(sensorPin);

  float voltage=value*5.0/1023;

  float temp=voltage*100;

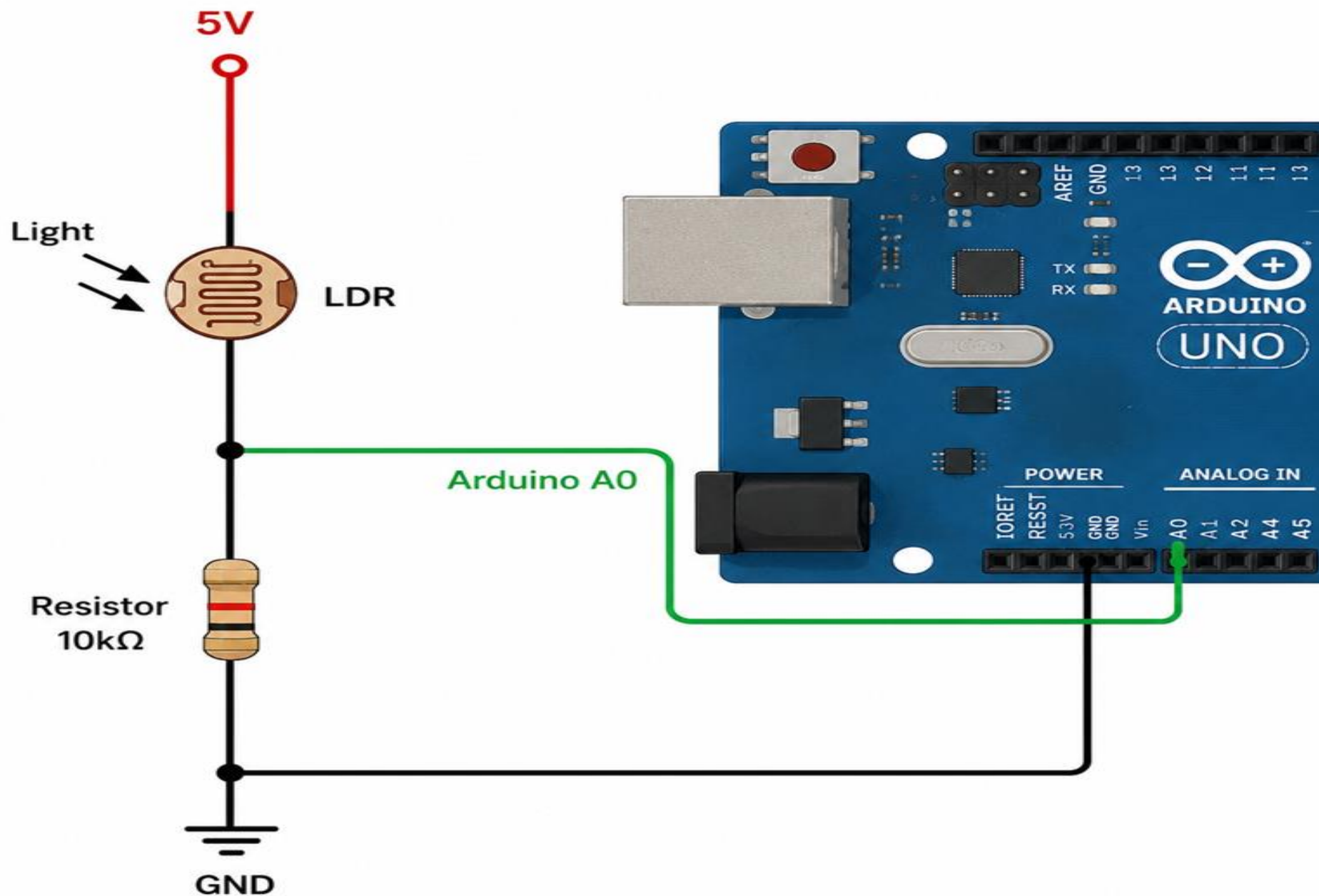
  Serial.println(temp);

  delay(1000);
}
```

Light Sensors

- **Light Dependent Resistor (LDR)**
 - Resistance changes with light intensity.
 - Applications:
 - Automatic streetlights
 - Solar trackers
 - Smart agriculture
 - **Operation**
 - **Bright Light:**
 - Low resistance
 - **Darkness:**
 - High resistance

LDR Circuit



Infrared Distance Sensors

IR distance sensor does distance or proximity sensing through emitting IR wave and calculating the angle of reflection.

IR sensors come with two lenses:

- An IR LED emitter lens that emits a light beam
- A position-sensible photodetector (PSD) where the reflected beam will fall onto

IR Distance Sensors: Working Principle

IR distance sensors work through the principle of triangulation; measuring distance based on the angle of the reflected beam.

Common infrared sensors



JSUMO
www.jsumo.com



Key Applications for IR Sensors

- TVs, computers, laptops
- Distance measurement
- Security systems such as surveillance, burglar alarms, etc.
- Monitoring and control applications

Advantages and Disadvantages of IR Sensors

Advantages

- Small form factor; Common IR sensors are smaller in size
- Applicable for daytime and nighttime usages
- Secured communication through a line of sight
- Able to measure the distance of objects that have complex surfaces unlike ultrasonic sensors

Disadvantages of IR sensors

- Limited measurement range
- Affected by environmental conditions and hard object

Sonar range finders (Ultrasonic sensors)

- Ultrasonic sensors Accurately measure distances over a wide range



Ultrasonic sensor (cont)

Wiring

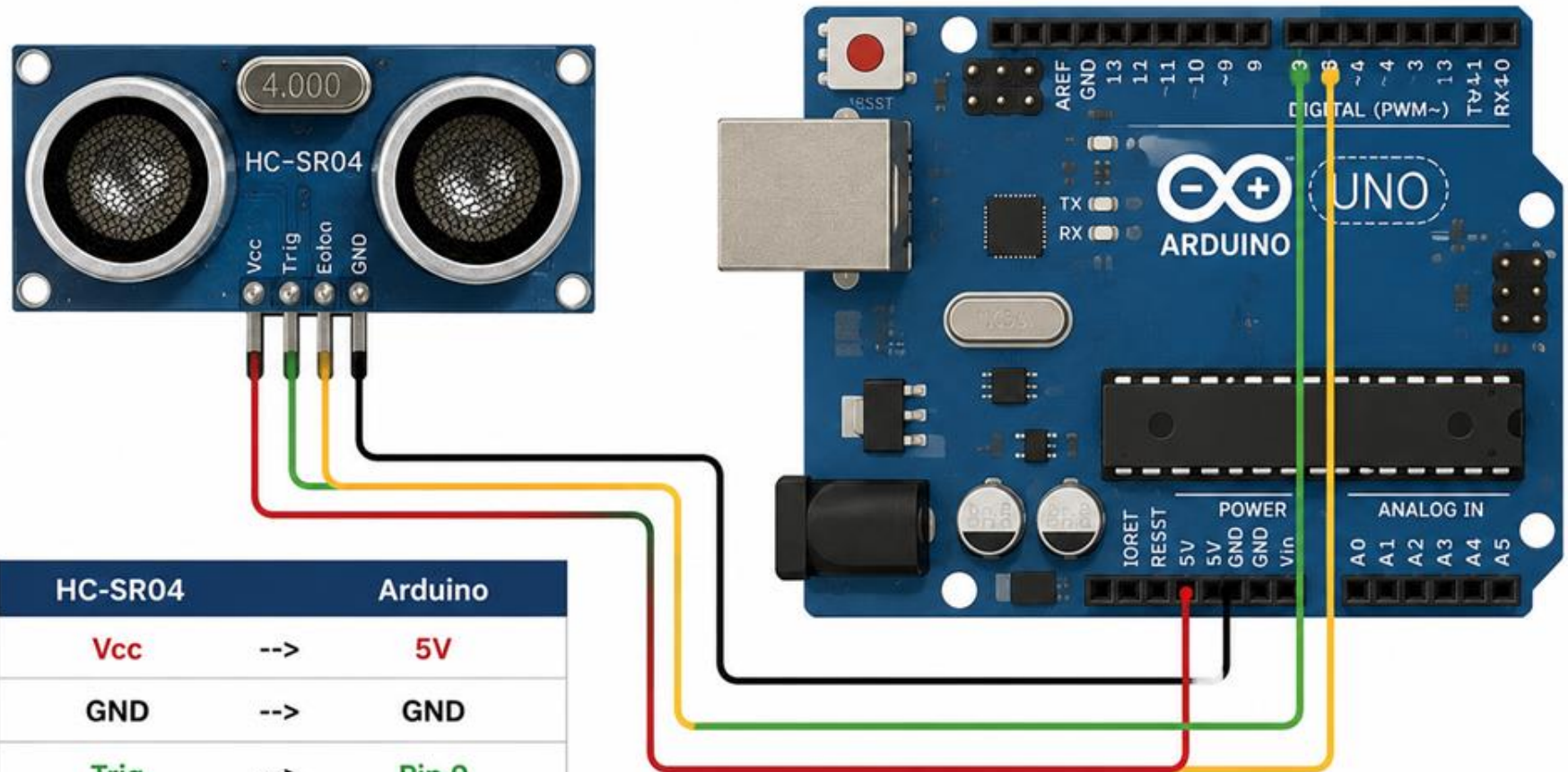
- 5V Supply
- Trigger Pulse Input
- Echo Pulse Output
- 0V Ground



Electric Parameter

Working Voltage	DC 5 V
Working Current	15mA
Working Frequency	40Hz
Max Range	4m
Min Range	2cm
MeasuringAngle	15 degree
Trigger Input Signal	10uS TTL pulse
Echo Output Signal	Input TTL lever signal and the range in proportion
Dimension	45*20*15mm

HC-SR04 Connections with Arduino



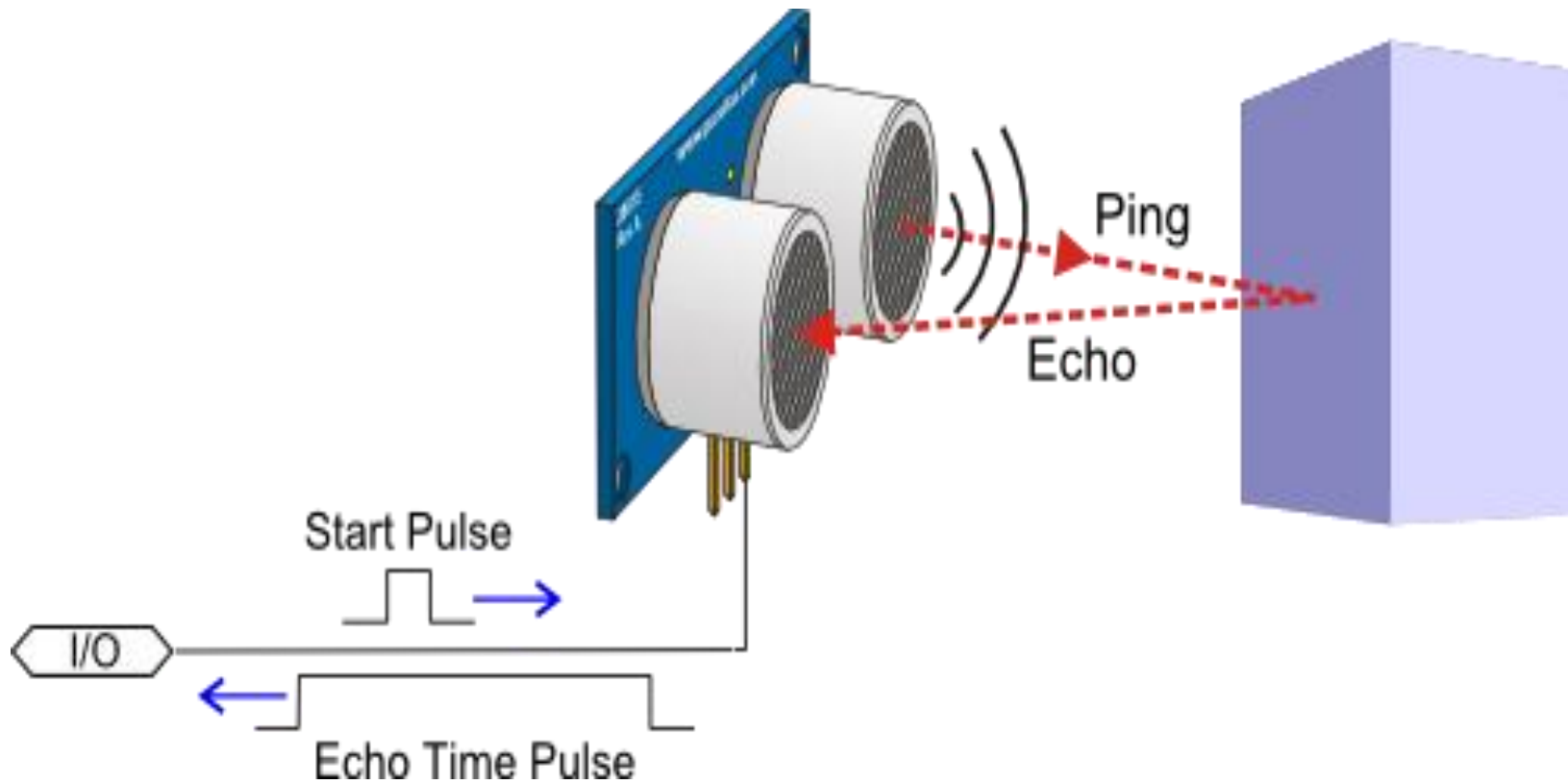
HC-SR04		Arduino	
Vcc	-->	5V	
GND	-->	GND	
Trig	-->	Pin 9	
Echo	-->	Pin 10	



Note:

Make sure to connect GND of the sensor and Arduino together.

How Ultrasonic sensor works



Calculating distance using Ultrasonic sensor

Distance = (Time x SpeedOfSound) / 2. The "2" is in the formula because the sound has to travel back and forth.

Speed of sound at sea level = 343 m/s
or 34300 cm/s

Sound speed and time given :-

Sound speed = 340 m/s = 0.034 cm/ μ s

$$\text{distance} = \frac{\text{speed} * \text{time}}{2}$$

Key applications of Ultrasonic sensors

- Distance Measurement
- Robotic Sensors
- Smart cars –Tesla uses ultrasonic sensors as part of its Autopilot program
- Unmanned Aerial Vehicles (UAV) / Drones

Advantages and disadvantages of Ultrasonic sensors

Advantages of Ultrasonic Sensors

- Not affected by object color and transparency as it detects distance through sound waves
- Works well in places that are dim
- Tend to consume lower current/power
- Multiple interface options for pairing with a microcontroller, etc.

Disadvantages of Ultrasonic Sensors

- Limited detection range
- Low resolution and slow refresh rate, making it not suitable for detection of fast-moving targets
- Unable to measure the distance of objects that have extreme textures/surfaces

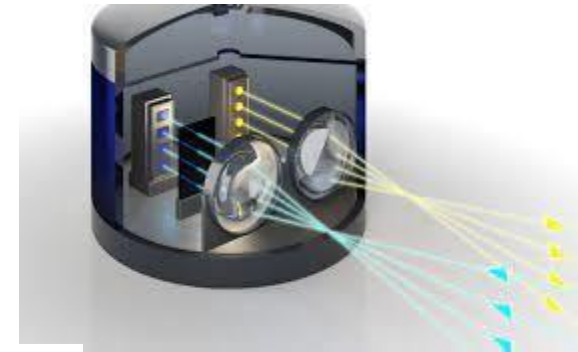
IR vrs Ultrasonic sensor

Type	IR Sensor	Ultrasonic Sensor
What It Does	Measure distance through reflected light waves	Measure distance through reflected sound waves
How It Measures	Triangulation: Angle of a reflected IR beam is measured	Time taken between transmitting and receiving sound waves are recorded
Human Interactions	Invisible to the naked eye	Unhearable
Object Requirements	Suitable to be used to measure complex objects	Not suitable to measure objects with complex surfaces

LiDAR Sensor

- ❖ Lidar stand for “**Light Detection And Ranging.**”
- ❖ It is sometimes called “**laser scanning**” or “**3D scanning.**”
- ❖ The technology uses eye-safe laser beams to create a 3D representation of the surveyed environment.
- ❖ It measures the range of targets through **light waves** from a laser instead of radio or sound waves.

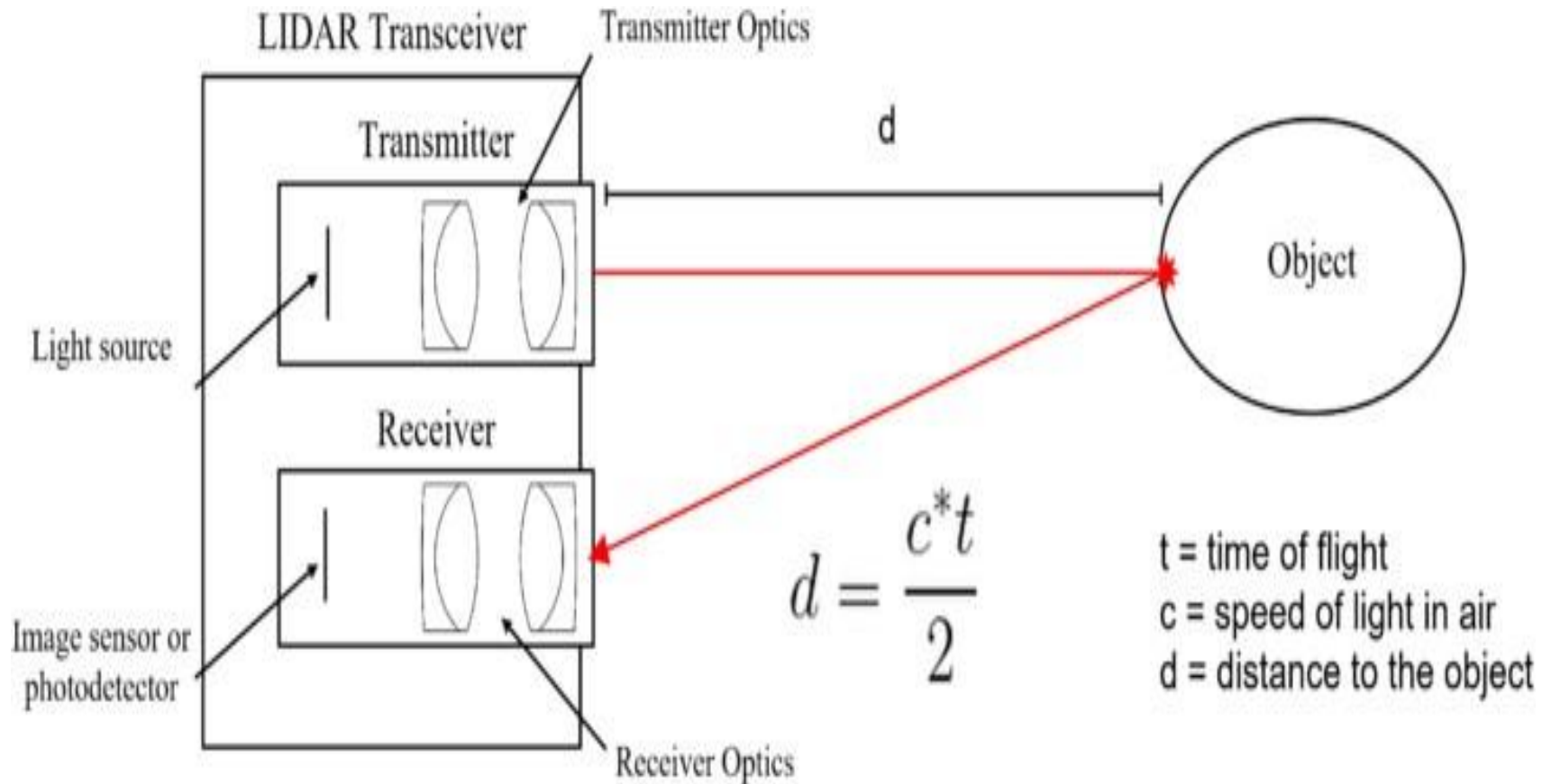
Sample LiDAR sensors



LiDAR : Working Principle

- A typical lidar sensor **emits pulsed light waves into the surrounding environment.**
- These pulses bounce off surrounding objects and return to the sensor. The sensor uses the time it took for each pulse to return to the sensor to calculate the distance it traveled.
- Repeating this process millions of times per second creates a precise, real-time 3D map of the environment. A robot can utilize this map for safe navigation.

LiDAR : Working Principle (cont)



LiDAR : Working Principle (cont)

1. The transmitter on the LiDAR device emits laser light at the target object
2. The pulse of the laser is reflected by the target object
3. Distance is then calculated by using the relationship between constant speed of light in air and the time between sending/receiving of the signal

Key Applications of LiDAR

- Environmental Monitoring; forestry, land mapping, etc.
- Distance Measurement
- Machine Control and Safety
- Robotics Imaging & Environmental Detection

Advantages and Disadvantages of LiDAR

Advantages of LiDAR

- High measurement range and accuracy
- Ability to measure 3D structures
- Fast update rate; suitable for fast-moving objects
- Small wavelengths as compared to sonar and radar; good at detecting small objects
- Applicable for usage in the day and night

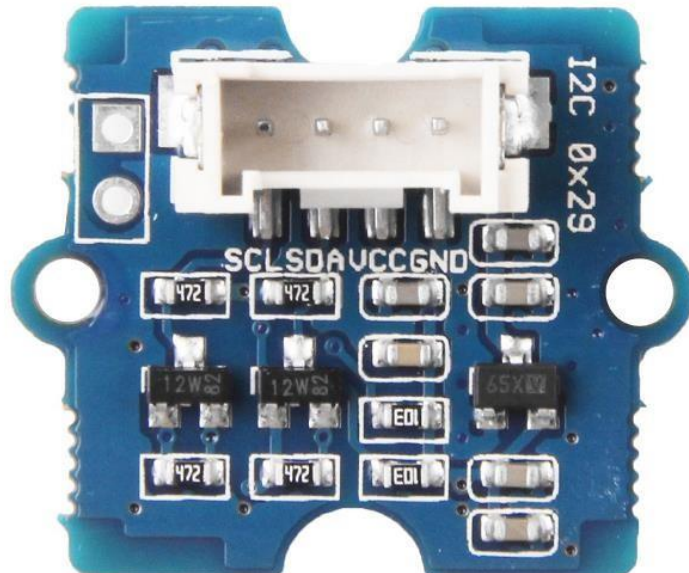
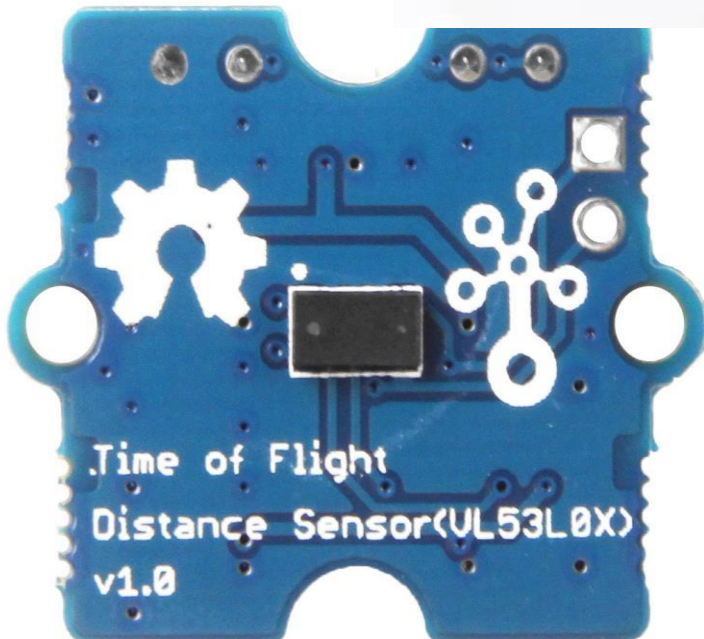
Disadvantages of LiDAR

- Higher cost as compared to ultrasonic and IR
- Harmful to the naked eye; higher-end LiDAR devices may utilize stronger LiDAR pulses which may affect the human eye

LED Time-Of-Flight Distance Sensors

- ❖ The Time-of-Flight principle (ToF) is a method for **measuring** the distance between a sensor and an object, based on the time difference between the emission of a signal and its return to the sensor, after being reflected by an object

Time of Flight Distance Sensors



Time-of-Flight Sensors: Working Principle

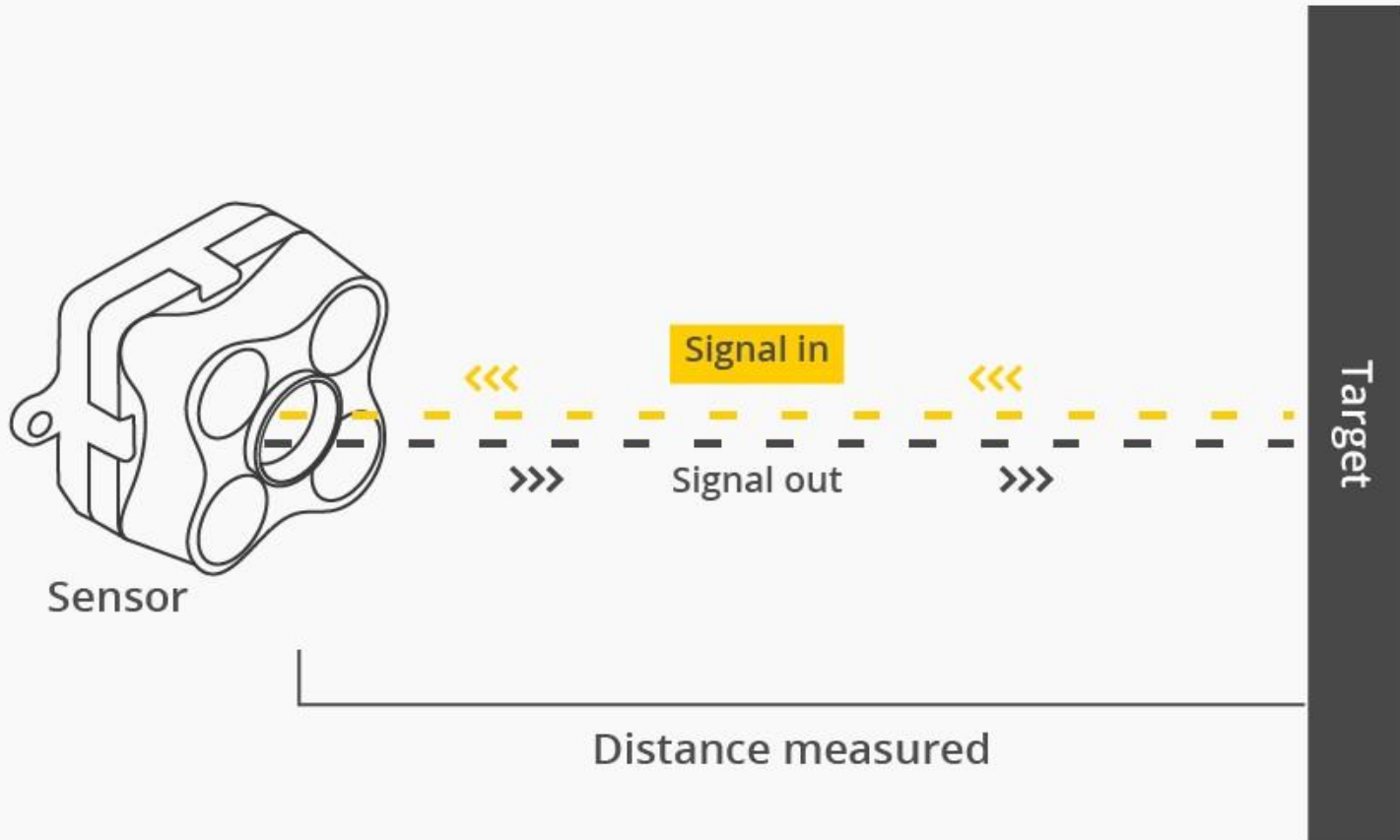
Time-of-Flight sensors measure the elapsed time from the emission of a **wave pulse** from the sensor to the moment it returns to the sensor after reflecting off an object.

It is capable of producing a 3D X, Y, Z image with a single snapshot by measuring the time it takes for light to travel from emitter to the receiver.

With time-of-flight technology, it provides significant benefits over other distance sensing methods (IR, Ultrasonic)

- Wider range
- Faster readings
- Greater accuracy

Time-of-Flight Sensors: Working Principle (cont)



Time-of-Flight Sensors: Working Principle (cont)

Time-of-flight sensors work similarly to LiDAR sensors, where:

1. The transmitter on the time-of-flight device emits **IR waves** towards the target object
2. The wave is reflected back upon reaching the target object
3. Distance is then calculated by using the speed of light in air and the time between sending/receiving of the signal

Key Applications of Time-of-Flight Sensors

- Industrial applications
- Machine vision
- Robotics
- People counting
- Drones

Advantages and Disadvantages

Advantages of Time-of-Flight Sensors

- Such technology offers high measurement range with accuracy
- 3D imaging capable
- Used in a wide variety of applications due to its ability to identify large objects

Disadvantages of Time-of-Flight Sensors

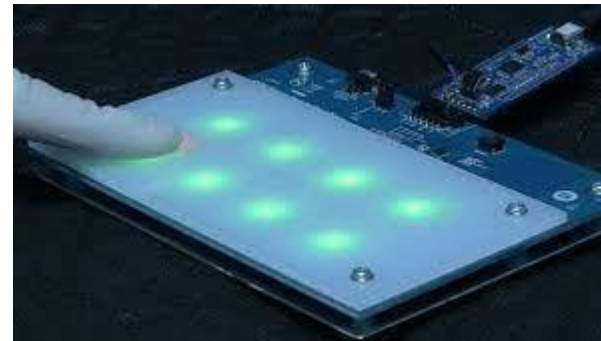
- Higher costs in general
- Z-depth resolution is still poor with general systems offering a 1cm Z-resolution

Comparing Range sensors (Ultrasonic, IR, LIDAR, ToF)

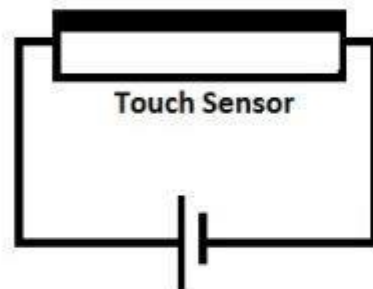
Type	Ultrasonic	IR	LIDAR	ToF
Suitablility for Long Range Sensing	No	No	Yes	Yes
High reading frequency	No	No	Yes	Yes
Cost	Low	Low	High	Moderate
Suitability to use for complex objects	No	Yes	Yes	Yes
Sensitive to external conditions	Yes	No	No	No
3D imaging compatible	No	No	Yes	Yes

Touch Sensors

- A touch sensor is a **type of device that captures and records physical touch or embrace on a device and/or object.**
- It enables a device or object to detect touch or near proximity, typically by a human user or operator are sensitive to touch, force or pressure.
- Touch sensors are used simply to indicate whether contact has been made with an object.

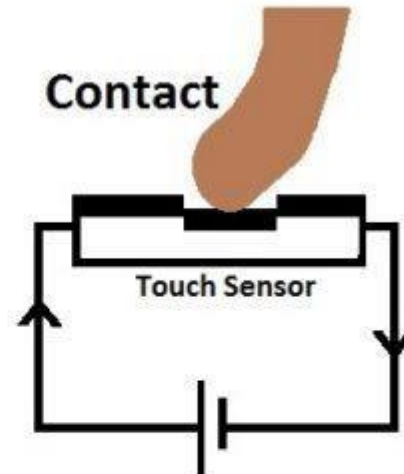


No contact



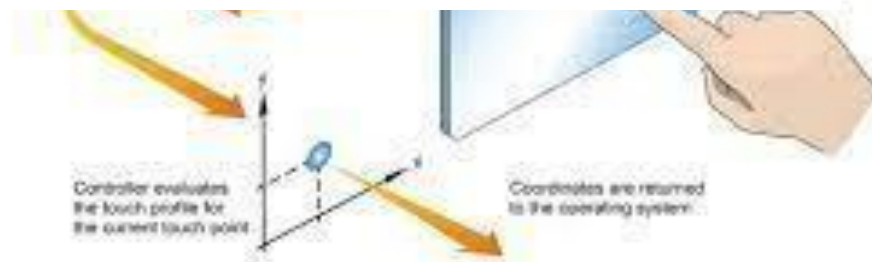
No Flow of current

Contact



Flow of current

Working Principle of TOUCH SENSOR by EEE Project

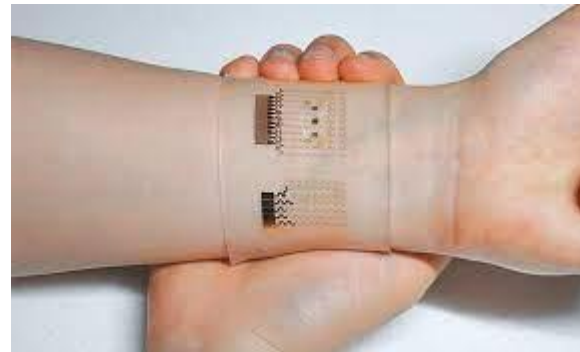


Applications of Touch sensors

- **Touch sensor in robotics;** a touch sensor is commonly used in robots, enabling basic movement and the ability to detect touch in its surroundings (E.g. When the robot runs into something, the touch sensor can have it to stop moving)
- Smartphones, automotive, industrial applications
- Touch sensor faucet in kitchens; allowing for control of running water without having to physically turn the knob
- Most other applications that require pressure/distance measurement

Stress sensors

Stress sensors are used to measure the magnitude of the contact force



- **Arduino Program**

```
const int trig=9;  
const int echo=10;
```

```
void setup()  
{  
  Serial.begin(9600);
```

```
  pinMode(trig,OUTPUT);  
  pinMode(echo,INPUT);  
}
```

```
void loop()  
{  
  digitalWrite(trig,LOW);  
  delayMicroseconds(2);  
  
  digitalWrite(trig,HIGH);  
  delayMicroseconds(10);  
  
  digitalWrite(trig,LOW);  
  
  long duration=pulseIn(echo,HIGH);  
  float distance=duration*0.0343/2;  
  Serial.println(distance);  
  delay(100);  
}
```

Assignment

- Design a temperature monitoring system using LM35.